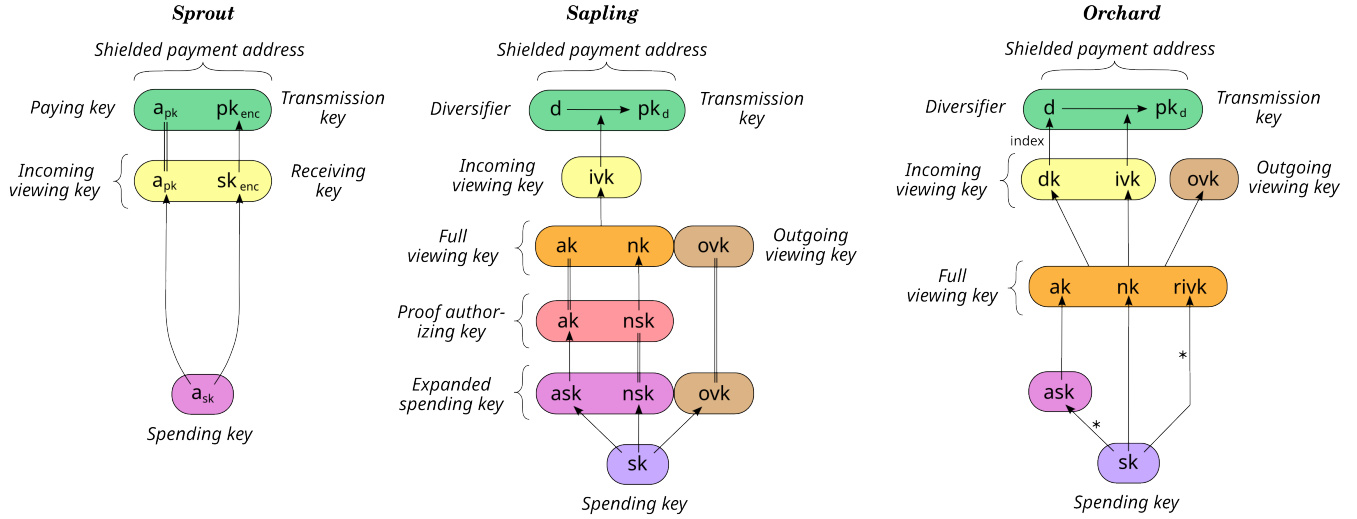


3 Concepts

3.1 Payment Addresses and Keys

Users who wish to receive shielded payments in the **Zcash** protocol must have a *shielded payment address*, which is generated from a *spending key*.

The following diagrams depict the relations between key components in **Sprout** and **Sapling** and **Orchard**. Arrows point from a component to any other component(s) that can be derived from it. Double lines indicate that the same component is used in multiple abstractions.



[Sprout] A Sprout receiving key sk_{enc} , incoming viewing key $ivk = (a_{pk}, sk_{enc})$, and shielded payment address $addr_{pk} = (a_{pk}, pk_{enc})$ are derived from the spending key a_{sk} , as described in §4.2.1 ‘*Sprout Key Components*’ on p. 38.

[Sapling onward] A Sapling expanded spending key is composed of a Spend authorizing key ask , a nullifier private key nsk , and an outgoing viewing key ovk . From these components we can derive a proof authorizing key (ak, nsk) , a full viewing key (ak, nk, ovk) , an incoming viewing key ivk , and a set of diversified payment addresses $addr_d = (d, pk_d)$, as described in §4.2.2 ‘*Sapling Key Components*’ on p. 38.

The consensus protocol does not depend on how an expanded spending key is constructed. Two methods of doing so are defined:

1. Generate a spending key sk at random and derive the expanded spending key (ask, nsk, ovk) from it, as shown in the diagram above and described in §4.2.2 ‘*Sapling Key Components*’ on p. 38.
2. Obtain an extended spending key as specified in [ZIP-32]; this includes a superset of the components of an expanded spending key. This method is used in the context of a *Hierarchical Deterministic Wallet*.

[NU5 onward] An Orchard spending key sk is used to derive a Spend authorizing key ask , and a full viewing key $(ak, nk, rinvk)$. From the full viewing key we can also derive an incoming viewing key (composed of a diversifier key dk and a $KA^{Orchard}$ private key ivk), an outgoing viewing key ovk , and a set of diversified payment addresses $addr_d = (d, pk_d)$, as described in §4.2.3 ‘*Orchard Key Components*’ on p. 40.

* The derivations of ask and $rinvk$ shown are not the only possibility. For further detail see §4.2.3 on p. 40.

Non-normative note: Most Zcash wallets derive Sapling and Orchard keys and addresses according to [ZIP-32].